

ARP与IP包分析

找到一个ARP请求分组，和对它的响应分组，观察ARP包的封装层次，是否所有的ARP请求都是广播包？（尝试使用显示过滤规则）

arp.pcapng

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

arp

No.	Time	Source	Destination	Protocol	Length	Info
245	18.449441	Intel_1a:bf:6c	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.9
249	18.755916	zte_c1:2a:f3	CloudNetwork_31:62:...	ARP	42	Who has 192.168.1.13? Tell 192.168.1.1
250	18.755983	CloudNetwork_31:62:...	zte_c1:2a:f3	ARP	42	192.168.1.13 is at dc:e9:94:31:62:63
251	18.961521	Intel_1a:bf:6c	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.9
411	47.325626	zte_c1:2a:f3	CloudNetwork_31:62:...	ARP	42	Who has 192.168.1.13? Tell 192.168.1.1
412	47.325680	CloudNetwork_31:62:...	zte_c1:2a:f3	ARP	42	192.168.1.13 is at dc:e9:94:31:62:63
446	53.784699	CloudNetwork_31:62:...	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.13
447	53.786287	zte_c1:2a:f3	CloudNetwork_31:62:...	ARP	42	192.168.1.1 is at 78:90:a2:c1:2a:f3
694	72.526343	zte_c1:2a:f3	CloudNetwork_31:62:...	ARP	42	Who has 192.168.1.13? Tell 192.168.1.1
695	72.526377	CloudNetwork_31:62:...	zte_c1:2a:f3	ARP	42	192.168.1.13 is at dc:e9:94:31:62:63
856	88.900895	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.100? Tell 192.168.1.17
857	88.901105	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.252? Tell 192.168.1.17
858	88.901105	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.114? Tell 192.168.1.17
859	89.105786	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.105? Tell 192.168.1.17
860	89.515165	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.100? Tell 192.168.1.17
861	89.515374	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.252? Tell 192.168.1.17
862	89.617635	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.114? Tell 192.168.1.17
864	90.539318	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.100? Tell 192.168.1.17
865	90.539512	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.252? Tell 192.168.1.17
866	90.539512	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.114? Tell 192.168.1.17
872	93.510197	Intel_70:f5:94	CloudNetwork_31:62:...	ARP	42	Who has 192.168.1.13? Tell 192.168.1.17
873	93.510220	CloudNetwork_31:62:...	Intel_70:f5:94	ARP	42	192.168.1.13 is at dc:e9:94:31:62:63
881	98.486564	zte_c1:2a:f3	CloudNetwork_31:62:...	ARP	42	Who has 192.168.1.13? Tell 192.168.1.1
882	98.486590	CloudNetwork_31:62:...	zte_c1:2a:f3	ARP	42	192.168.1.13 is at dc:e9:94:31:62:63
885	99.960051	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.17
888	100.881664	Intel_70:f5:94	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.17

Address Resolution Protocol: Protocol

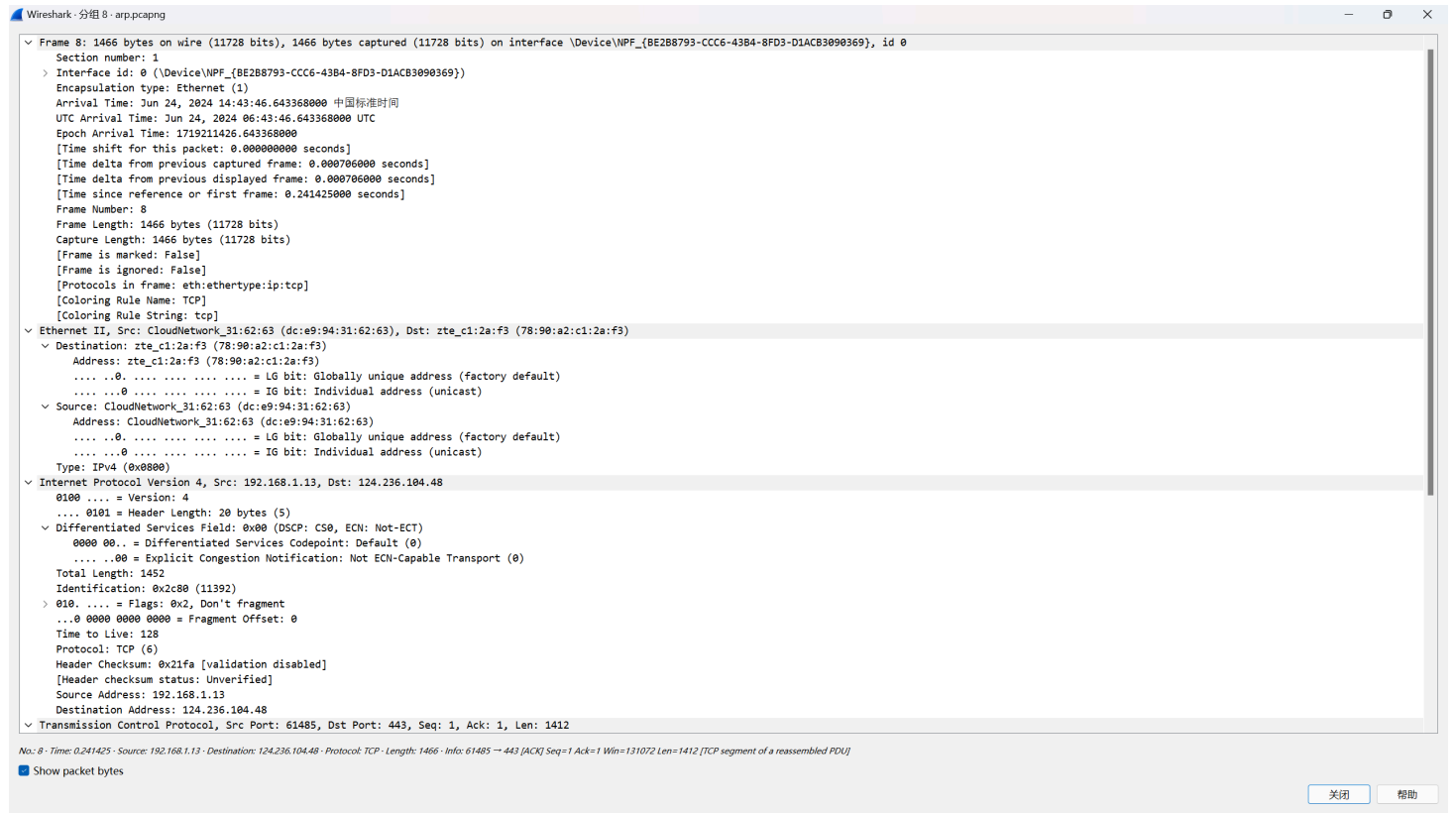
分组: 925 · 已显示 26 (2.8%)

配置: Default

不是所有ARP请求都是广播包

找到一个IPv4报文，观察IP层首部，分析IP首部各字段值，并说明其含义，例如该IP协议版本号是多少？首部长

度是几个字节？整个IP数据报长度是多少字节？源IP地址和目的IP地址各是什么？等等,尽量全面。



ip协议版本号是4
首部长度是20字节
整个长度是1452字节
源IP是192.168.1.13
目的ip是124.236.104.48

根据抓到的IP Header内容，计算校验和，并验证。

首部校验和=ffff-(4500+05ac+2c80+4000+8006+21fa+c0a8+010d+7cec+6830)=0x21fa
相同